

1.4 Systems of equations and consistency

A system $\mathbf{Ax} = \mathbf{b}$ with $\mathbf{A}$ of size $m \times n$ is **consistent** if it has at least one solution. Least squares leads to such systems in which $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$ may be singular, so we need to know when solutions exist, what all of them look like, and which features are common to all of them.

Theorem 1.4.1 — Consistency and the general solution

Let $\mathbf{G}$ be any generalized inverse of $\mathbf{A}$. The following are equivalent:

- (a) $\mathbf{Ax} = \mathbf{b}$ is consistent;
- (b) $\mathbf{b} \in \mathcal{C}(\mathbf{A})$;
- (c) $\text{rank}[\mathbf{A}, \mathbf{b}] = \text{rank}(\mathbf{A})$;
- (d) $\mathbf{AGb} = \mathbf{b}$.

If the system is consistent, $\mathbf{x}_0 = \mathbf{Gb}$ is a solution, and the set of all solutions is

$$\{\mathbf{Gb} + (\mathbf{I} - \mathbf{GA})\mathbf{z} : \mathbf{z} \in \mathbb{R}^n\} = \mathbf{x}_0 + \mathcal{N}(\mathbf{A}). \quad (1.4.1)$$

Proof. (a) $\Leftrightarrow$ (b) is the first reading in Eq. 1.1.2. (b) $\Leftrightarrow$ (c): appending $\mathbf{b}$ as a column leaves the column space unchanged iff $\mathbf{b}$ is already in it, and a subspace of $\mathcal{C}([\mathbf{A}, \mathbf{b}])$ with the same dimension equals it. (d) $\Rightarrow$ (a): $\mathbf{Gb}$ is a solution. (b) $\Rightarrow$ (d): if $\mathbf{b} = \mathbf{Au}$, then $\mathbf{AGb} = \mathbf{AGA}\mathbf{u} = \mathbf{Au} = \mathbf{b}$.

The solutions of a consistent system are $\mathbf{x}_0 + \mathcal{N}(\mathbf{A})$, since $\mathbf{Ax} = \mathbf{b}$ iff $\mathbf{A}(\mathbf{x} - \mathbf{x}_0) = \mathbf{0}$. It remains to show $\mathcal{N}(\mathbf{A}) = \mathcal{C}(\mathbf{I} - \mathbf{GA})$. The inclusion $\supseteq$ holds because $\mathbf{A}(\mathbf{I} - \mathbf{GA}) = \mathbf{0}$. Conversely, if $\mathbf{Av} = \mathbf{0}$, then $\mathbf{v} = (\mathbf{I} - \mathbf{GA})\mathbf{v}$.

■

Criterion (d) is useful in practice: it tests consistency with a single matrix product, for any generalized inverse. By Proposition 1.3.3(a) and rank-nullity, $\text{rank}(\mathbf{I} - \mathbf{GA}) = n - r$, so the solution set is an affine subspace of dimension $n - r$. The solution is unique iff $\text{rank}(\mathbf{A}) = n$, and then $\mathbf{Gb}$ is the same for every $\mathbf{G}$.

Idea. Any generalized inverse does three jobs for $\mathbf{Ax} = \mathbf{b}$. It tests consistency ($\mathbf{AGb} = \mathbf{b}$), produces one solution ($\mathbf{Gb}$), and parametrizes all of them ($\mathbf{Gb} + (\mathbf{I} - \mathbf{GA})\mathbf{z}$). Which $\mathbf{G}$ is used changes the particular solution but never the solution set, and a linear function $\mathbf{q}^\top \mathbf{x}$ is the same across solutions exactly when $\mathbf{q} \in \mathcal{C}(\mathbf{A}^\top)$.

Corollary 1.4.2 — Invariant linear functions

Let $\mathbf{Ax} = \mathbf{b}$ be consistent and let $\mathbf{q} \in \mathbb{R}^n$. The value $\mathbf{q}^\top \mathbf{x}$ is the same for every solution $\mathbf{x}$ iff $\mathbf{q} \in \mathcal{C}(\mathbf{A}^\top)$, that is, iff $\mathbf{q}^\top = \mathbf{q}^\top \mathbf{GA}$ for one (and then every) generalized inverse $\mathbf{G}$.

Proof. By Eq. 1.4.1, $\mathbf{q}^\top \mathbf{x}$ is constant over solutions iff $\mathbf{q}^\top (\mathbf{I} - \mathbf{GA})\mathbf{z} = 0$ for all $\mathbf{z}$, that is, iff $\mathbf{q}^\top = \mathbf{q}^\top \mathbf{GA}$. That equation puts $\mathbf{q}$ in $\mathcal{C}(\mathbf{A}^\top)$. Conversely, if $\mathbf{q} = \mathbf{A}^\top \mathbf{w}$, then $\mathbf{q}^\top \mathbf{GA} = \mathbf{w}^\top \mathbf{AGA} = \mathbf{w}^\top \mathbf{A} = \mathbf{q}^\top$. ■

Applied to the normal equations $\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{y}$, which are consistent by Proposition 1.2.3(c), the corollary says that $\mathbf{q}^\top \hat{\boldsymbol{\beta}}$ is the same for all least squares solutions iff $\mathbf{q} \in \mathcal{C}(\mathbf{X}^\top \mathbf{X}) = \mathcal{C}(\mathbf{X}^\top)$. This is the algebra behind estimable functions (Theorem 8.3.1).

Proposition 1.4.3 — Minimum-norm solution

If $\mathbf{Ax} = \mathbf{b}$ is consistent, then $\mathbf{A}^+ \mathbf{b}$ is its unique solution lying in $\mathcal{C}(\mathbf{A}^\top)$, and $\|\mathbf{A}^+ \mathbf{b}\| < \|\mathbf{x}\|$ for every other solution $\mathbf{x}$.

Proof. $\mathbf{A}^+ \mathbf{b}$ is a solution because $\mathbf{A}^+$ is a generalized inverse. From the second and fourth Penrose conditions, $\mathbf{A}^+ = (\mathbf{A}^+ \mathbf{A}) \mathbf{A}^+ = (\mathbf{A}^+ \mathbf{A})^\top \mathbf{A}^+ = \mathbf{A}^\top (\mathbf{A}^+)^{\top} \mathbf{A}^+$, so $\mathbf{A}^+ \mathbf{b} \in \mathcal{C}(\mathbf{A}^\top)$. Every vector of $\mathcal{C}(\mathbf{A}^\top)$ is orthogonal to every $\mathbf{v} \in \mathcal{N}(\mathbf{A})$, since $(\mathbf{A}^\top \mathbf{w})^\top \mathbf{v} = \mathbf{w}^\top \mathbf{Av} = 0$. Any other solution is $\mathbf{x} = \mathbf{A}^+ \mathbf{b} + \mathbf{v}$ with $\mathbf{0} \neq \mathbf{v} \in \mathcal{N}(\mathbf{A})$, and then $\|\mathbf{x}\|^2 = \|\mathbf{A}^+ \mathbf{b}\|^2 + \|\mathbf{v}\|^2$. The same computation shows that $\mathbf{x} \notin \mathcal{C}(\mathbf{A}^\top)$, because $\mathbf{x}^\top \mathbf{v} = \|\mathbf{v}\|^2 \neq 0$. ■

Example. (Filling a table from its margins)

Suppose the six entries of a 2×3 table are unknown, but its row totals (9, 6) and column totals (4, 5, 6) are known. With the cells ordered $x_{11}, x_{12}, x_{13}, x_{21}, x_{22}, x_{23}$, the equations are $\mathbf{Ax} = \mathbf{b}$ with

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 9 \\ 6 \\ 4 \\ 5 \\ 6 \end{pmatrix}.$$

The rows of $\mathbf{A}$ satisfy one relation: the two row-total rows and the three column-total rows both add to $\mathbf{1}_6^\top$. So $\text{rank}(\mathbf{A}) = 4$ and there is a $6 - 4 = 2$ -dimensional family of solutions. The same relation must hold for $\mathbf{b}$. By Theorem 1.4.1(c), the system is consistent iff the row totals and the column totals have the same sum, and here both sums are 15.

The listing builds three generalized inverses: $\mathbf{G}_1$ from the nonsingular submatrix in rows 1–4 and columns 1–4 (Theorem 1.3.2), the Moore–Penrose inverse $\mathbf{G}_2 = \mathbf{A}^+$, and $\mathbf{G}_3 = \mathbf{G}_1 + (\mathbf{I} - \mathbf{G}_1 \mathbf{A}) \mathbf{Z}$ for an integer matrix $\mathbf{Z}$ (Proposition 1.3.3(e)). They give the solutions

$$\begin{aligned} \mathbf{G}_1 \mathbf{b} &= (-2, 5, 6, 6, 0, 0)^\top, & \|\mathbf{G}_1 \mathbf{b}\| &= 10.050, \\ \mathbf{G}_2 \mathbf{b} &= (5/2, 3, 7/2, 3/2, 2, 5/2)^\top, & \|\mathbf{G}_2 \mathbf{b}\| &= 6.325, \\ \mathbf{G}_3 \mathbf{b} &= (21, -39, 27, -17, 44, -21)^\top, & \|\mathbf{G}_3 \mathbf{b}\| &= 73.192. \end{aligned}$$

The individual cell x_{11} takes the values $-2, 5/2$ and 21 : margins do not determine

cells. But $x_{12} + x_{13} - x_{21}$ equals 5 for all three, as [Corollary 1.4.2](#) predicts, because its coefficient vector is the first row of $\mathbf{A}$ minus the third. The minimum-norm solution has the additive form $x_{ij} = r_i/3 + c_j/2 - 15/6$ in terms of the row totals r_i and column totals c_j ([Exercise \(A2\)](#)). If the last column total is changed to 7, then $\mathbf{A}\mathbf{G}\mathbf{b} \neq \mathbf{b}$ for each of the three matrices, and the script confirms that the test detects the inconsistency. The script also checks all 60 submatrices formed by 4 independent rows and 4 independent columns, and finds each one nonsingular, as Proposition 1.2.4 requires.

```
from fractions import Fraction
from itertools import combinations
import numpy as np
m, n = A.shape
# cells ordered (1,1), (1,2), (1,3), (2,1), (2,2), (2,3)
A = np.array([[1, 1, 1, 0, 0, 0],      # row 1 total
              [0, 0, 0, 1, 1, 1],      # row 2 total
              [1, 0, 0, 1, 0, 0],      # column 1 total
              [0, 1, 0, 0, 1, 0],      # column 2 total
              [0, 0, 1, 0, 0, 1]],      # column 3 total
             dtype=float)
b = np.array([9, 6, 4, 5, 6], dtype=float)      # both sets of totals add to 15
r = np.linalg.matrix_rank(A)
print("rank(A) =", r, " rank([A, b]) =", np.linalg.matrix_rank(np.column_stack([A, b]
)))

def ginverse_from_submatrix(A, rows, cols):
    """Place the inverse of the nonsingular block A[rows, cols] at [cols, rows]."""
    G = np.zeros(A.T.shape)
    G[np.ix_(cols, rows)] = np.linalg.inv(A[np.ix_(rows, cols)])
    return G

rows, cols = [0, 1, 2, 3], [0, 1, 2, 3]          # r independent rows and
columns                                           columns
G1 = ginverse_from_submatrix(A, rows, cols)
G2 = np.linalg.pinv(A)                          # Moore-Penrose inverse
rng = np.random.default_rng(1)
Z = rng.integers(-2, 3, size=A.T.shape).astype(float)
G3 = G1 + (np.eye(n) - G1 @ A) @ Z              # another, non-reflexive
one

for G in (G1, G2, G3):
    assert np.allclose(A @ G @ A, A)             # each is a g-inverse
    assert np.allclose(A @ G @ b, b)             # consistency test passes
    x = G @ b
    print(np.round(x, 4), " A x - b =", np.abs(A @ x - b).max())
```

[Run this code online](#)

1.4.1 Exercises

1.4.1.1 A. Check your understanding

Exercise A1.

Find all generalized inverses of a nonzero vector $\mathbf{a} \in \mathbb{R}^n$, regarded as an $n \times 1$ matrix.

Which one is $\mathbf{a}^+$?

Exercise A2.

Generalize [Example \(Filling a table from its margins\)](#) to an $a \times b$ table with row totals r_i , column totals c_j and grand total $T = \sum_i r_i = \sum_j c_j$. Show that $x_{ij} = r_i/b + c_j/a - T/(ab)$ is a solution lying in $\mathcal{C}(\mathbf{A}^\top)$, and hence is the minimum-norm solution.

Solution. Row i of the proposed table sums to $r_i + \sum_j c_j/a - bT/(ab) = r_i$, and column j sums to $\sum_i r_i/b + c_j - T/b = c_j$. The rows of $\mathbf{A}$ are the row indicators and the column indicators, so $\mathcal{C}(\mathbf{A}^\top)$ consists of the tables of the form $x_{ij} = \alpha_i + \beta_j$. The proposed solution has this form. By [Proposition 1.4.3](#), it is $\mathbf{A}^+\mathbf{b}$. For $a = 2$, $b = 3$ and $T = 15$ this is the additive formula quoted in [Example \(Filling a table from its margins\)](#).

Exercise A3.

For which c and d is the system $x_1 + x_2 + x_3 = 2$, $x_1 - x_2 = c$, $3x_1 + x_2 + 2x_3 = d$ consistent? When it is, give its general solution in the form [Eq. 1.4.1](#).

1.4.1.2 B. Practice

Exercise B1.

Let $\mathbf{G}$ be a generalized inverse of $\mathbf{A}$. Show that $\mathbf{GAG} = \mathbf{G}$ iff $\text{rank}(\mathbf{G}) = \text{rank}(\mathbf{A})$.

Solution. If $\mathbf{GAG} = \mathbf{G}$, then $\text{rank}(\mathbf{G}) \leq \text{rank}(\mathbf{A})$ by [Proposition 1.2.3\(a\)](#), and $\text{rank}(\mathbf{A}) \leq \text{rank}(\mathbf{G})$ by [Proposition 1.3.3\(a\)](#). Conversely, suppose $\text{rank}(\mathbf{G}) = \text{rank}(\mathbf{A}) = r$. Since $\mathcal{C}(\mathbf{GA}) \subseteq \mathcal{C}(\mathbf{G})$ and $\text{rank}(\mathbf{GA}) = r$, the two column spaces coincide, so $\mathbf{G} = \mathbf{GAK}$ for some $\mathbf{K}$. Then $\mathbf{GAG} = \mathbf{GAGAK} = \mathbf{GAK} = \mathbf{G}$, using $\mathbf{AGA} = \mathbf{A}$.

Exercise B2.

Let $\mathbf{A} = \mathbf{BF}$ be a rank factorization. Show that $\mathbf{A}^+ = \mathbf{F}^\top(\mathbf{FF}^\top)^{-1}(\mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top$. Deduce that $\mathbf{A}^+ = (\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{A}^\top$ when $\mathbf{A}$ has full column rank.

Solution. Write $\mathbf{G}$ for the proposed matrix. Then $\mathbf{AG} = \mathbf{B}(\mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top$ and $\mathbf{GA} = \mathbf{F}^\top(\mathbf{FF}^\top)^{-1}\mathbf{F}$, both symmetric. Also $\mathbf{AGA} = \mathbf{B}(\mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top\mathbf{BF} = \mathbf{A}$ and $\mathbf{GAG} = \mathbf{F}^\top(\mathbf{FF}^\top)^{-1}\mathbf{FF}^\top(\mathbf{FF}^\top)^{-1}(\mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top = \mathbf{G}$. By uniqueness $\mathbf{G} = \mathbf{A}^+$. If $\mathbf{A}$ has full column rank, take $\mathbf{B} = \mathbf{A}$ and $\mathbf{F} = \mathbf{I}$.

Exercise B3.

Let $\mathbf{A}$ be $m \times n$ of rank r and let $\mathbf{Ax} = \mathbf{b}$ be consistent with $\mathbf{b} \neq \mathbf{0}$. Show that the system has $n - r + 1$ linearly independent solutions and no more, and that every solution is an affine combination of them.

1.4.1.3 C. Going deeper**Exercise C1.**

Show that $(\mathbf{A}^+)^+ = \mathbf{A}$, $(\mathbf{A}^\top)^+ = (\mathbf{A}^+)^\top$, $(\mathbf{A}^\top \mathbf{A})^+ = \mathbf{A}^+ (\mathbf{A}^+)^\top$ and $\mathbf{A}^+ = (\mathbf{A}^\top \mathbf{A})^+ \mathbf{A}^\top$.
Give an example with $(\mathbf{AB})^+ \neq \mathbf{B}^+ \mathbf{A}^+$.